$$\langle c^\dagger_{1\uparrow} c^\dagger_{2\downarrow} c_{3\downarrow} c_{4\uparrow} \rangle = \overbrace{c^\dagger_{1\uparrow}}\, \overbrace{c^\dagger_{2\downarrow}}\, \overbrace{c_{3\downarrow}}\, \overbrace{c_{4\uparrow}} + \overbrace{c^\dagger_{1\uparrow} \overbrace{c^\dagger_{2\downarrow}}\, c_{3\downarrow}}\, \overbrace{c_{4\uparrow}} = \Delta_{34} \Delta^*_{12} + \chi_{14} \chi_{23}$$